

Scientific Summary

Evolutionary Conservation of the Human TP53 Protein

The **TP53** gene encodes the **tumor protein p53**, a sequence-specific transcription factor that plays a central role in preserving genomic stability. Often referred to as the "**guardian of the genome**," p53 is activated in response to various cellular stresses, including DNA damage, oxidative stress, oncogene activation, hypoxia, and viral infection. Upon activation, it regulates the expression of numerous target genes involved in **cell-cycle arrest, DNA repair, apoptosis, senescence, and metabolic regulation**, thereby preventing the propagation of damaged cells and reducing the risk of tumor development.

Due to its indispensable biological functions, the TP53 protein is highly conserved among vertebrates, particularly within mammals. This evolutionary conservation reflects the strong selective pressure to maintain its structural integrity and functional domains, including the **transactivation domain, DNA-binding domain, oligomerization domain, and regulatory C-terminal domain**. Variations within these regions may influence protein function, making TP53 an important model for comparative evolutionary studies.

In this project, the **human TP53 protein sequence (RefSeq Accession: NP_000537.3)** serves as the reference sequence for identifying homologous TP53 proteins from multiple mammalian species using **BLASTp**. The retrieved protein sequences are aligned using the **ClustalW** algorithm in **MEGA 12** to identify conserved amino acid residues and functionally important regions. The aligned sequences are subsequently used to construct a **Neighbor-Joining phylogenetic tree**, providing insights into the evolutionary relationships among TP53 proteins across different species. The resulting phylogenetic tree is further visualized and annotated using **Interactive Tree of Life (iTOL)** to improve interpretation and presentation.

This project demonstrates a complete comparative bioinformatics workflow, integrating **sequence retrieval, homology searching, multiple sequence alignment, phylogenetic reconstruction, and tree visualization**. Through these analyses, participants gain practical experience in evolutionary bioinformatics while developing an understanding of how sequence conservation reflects the functional importance of proteins. The study also highlights the value of phylogenetic analysis in exploring molecular evolution, comparative genomics, and the conservation of essential genes and proteins across species.